PRODUCT INVENTORY & PERFORMANCE

BUSINESS ANALYTICS REPORT

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1. Introduction

Inventory analysis is important because a business needs to keep enough products available while also avoiding unnecessary money being tied up in stock. For this assessment, I used Microsoft Power BI to analyse the products.csv dataset. The dashboard combines product information with price, stock quantity, customer rating, warranty and expiry-related information.

The main purpose of the dashboard is to make the product data easier to understand. Instead of checking individual records, the user can first look at the overall KPIs and charts and then use the filters or drillthrough page to investigate a particular product or group.

2. Objectives

The main aim of this project is to develop an interactive dashboard that can support inventory and product-performance decisions. This follows the assessment requirement to analyse inventory, pricing, ratings, warranties and product categories.

- Prepare the product dataset for analysis in Power BI.
- Create calculated fields for Inventory Value, Product Age and Stock Status.
- Create the required DAX measures for the main KPIs.
- Compare products and categories using suitable Power BI visuals.
- Analyse stock levels and identify products that may need replenishment.
- Examine the relationship between product price and customer rating.
- Monitor warranty information and expiry trends.
- Use slicers and drill-through to make the report interactive.

3. Dataset and Data Preparation

The dataset used for this project is products.csv. The available information includes product ID, product name, product category, price, stock quantity, product rating, warranty period and colour/size variations. Inventory Value is calculated from price and stock quantity. The supplied report describes the dataset as containing roughly 10,000 products across Electronics, Clothing and Home Appliances, with a combined inventory value of about $128.83 million.

3.1 Important Fields

Field	Use in Analysis
Product ID	Identifies each product record.
Product Category	Used to compare Electronics, Clothing and Home Appliances.
Price	Used for price analysis and price-band grouping.
Stock Quantity	Used to identify stock levels and inventory size.
Product Rating	Used to study customer satisfaction.
Warranty Period	Used to compare warranty groups.
Colour/Size Variations	Used as an interactive filter.
Inventory Value	Calculated as Price × Stock Quantity.

3.2 Data Preparation

Before creating the visuals, the dataset needs to be checked for suitable data types and usable values. The calculated fields are then created so that the same definitions can be used throughout the dashboard. The assessment specifically lists Inventory Value, Product Age and Stock Status as calculated fields.

- Inventory Value = Price × Stock Quantity.
- Stock Status is used to separate Low and Normal stock.
- Product Age/expiry information is used for ageing-stock analysis.
- Price bands are used to compare the amount of money held in different price ranges.

4. Power BI Measures and Dashboard Design

Five main measures were used for the KPI section. These measures give a quick summary before the user moves to the detailed charts.

Measure	Purpose
Total Products	Shows the number of products in the dataset.
Total Inventory Value	Shows the total financial value of available stock.
Average Price	Shows the average selling price.
Average Rating	Shows the average customer rating.
Low Stock Products	Shows how many products are currently classified as low stock.

4.1 Dashboard Interactivity

The report uses Product Category, Product Name, Warranty Period and Colour/Size Variations as slicers. A drill-through product details page is also included. This means that the dashboard can be used first for an overall view and then for a more focused investigation.

4.2 Main KPI Results

- Total Products: 10K
- Total Inventory Value: $128.83M
- Average Rating: 3.00
- Average Price: $254.67
- Low Stock Products: 2K

These values are shown in the supplied overview dashboard.

5. Dashboard Results and Discussion

5.1 Overview Dashboard

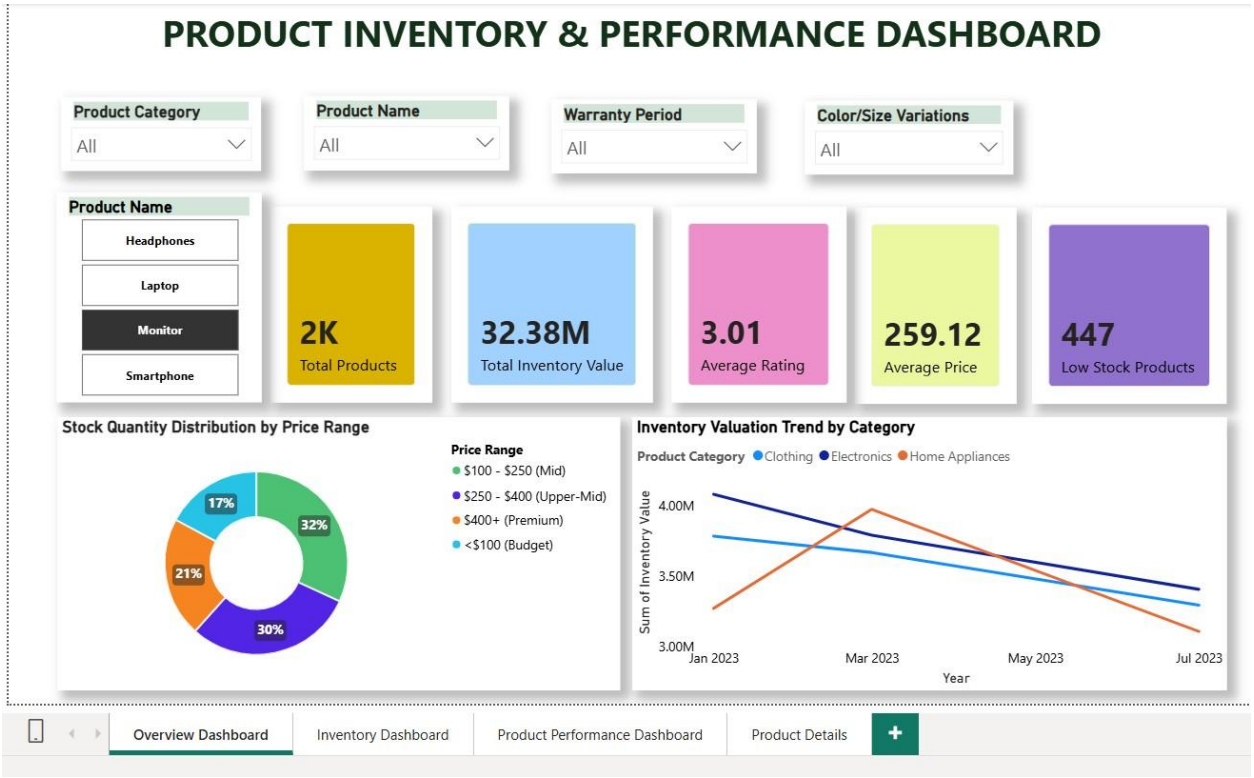


Figure 5.1 – Overview Dashboard

The overview page gives the main picture of the product catalogue. The five KPI cards are useful because they show the current size, value, price, rating and low-stock position without requiring the user to read the detailed charts first.

The most important point from this page is the scale of the inventory. With approximately $128.83M in inventory value, stock management is not only an operational issue; it is also a financial issue. The lowstock result of around 2K products also shows that replenishment needs to be monitored across the catalogue.

Note: Figure 5.1 is captured with the Product Name slicer set to "Monitor" to demonstrate the filtering feature, so the KPI cards in the screenshot (2K Total Products, $32.38M Inventory Value) show the filtered subset rather than the full catalogue. The unfiltered totals used throughout this report (10K products, $128.83M inventory value) are taken from the default, unfiltered view shown in Figure 8.1.

5.2 Why the Overview is Useful

- It provides a quick starting point for management.
- It makes the major KPIs visible in one place.
- It allows the user to move from summary information to detailed pages.
- It provides context for the category and inventory charts used later in the report. **6. Inventory Analysis**

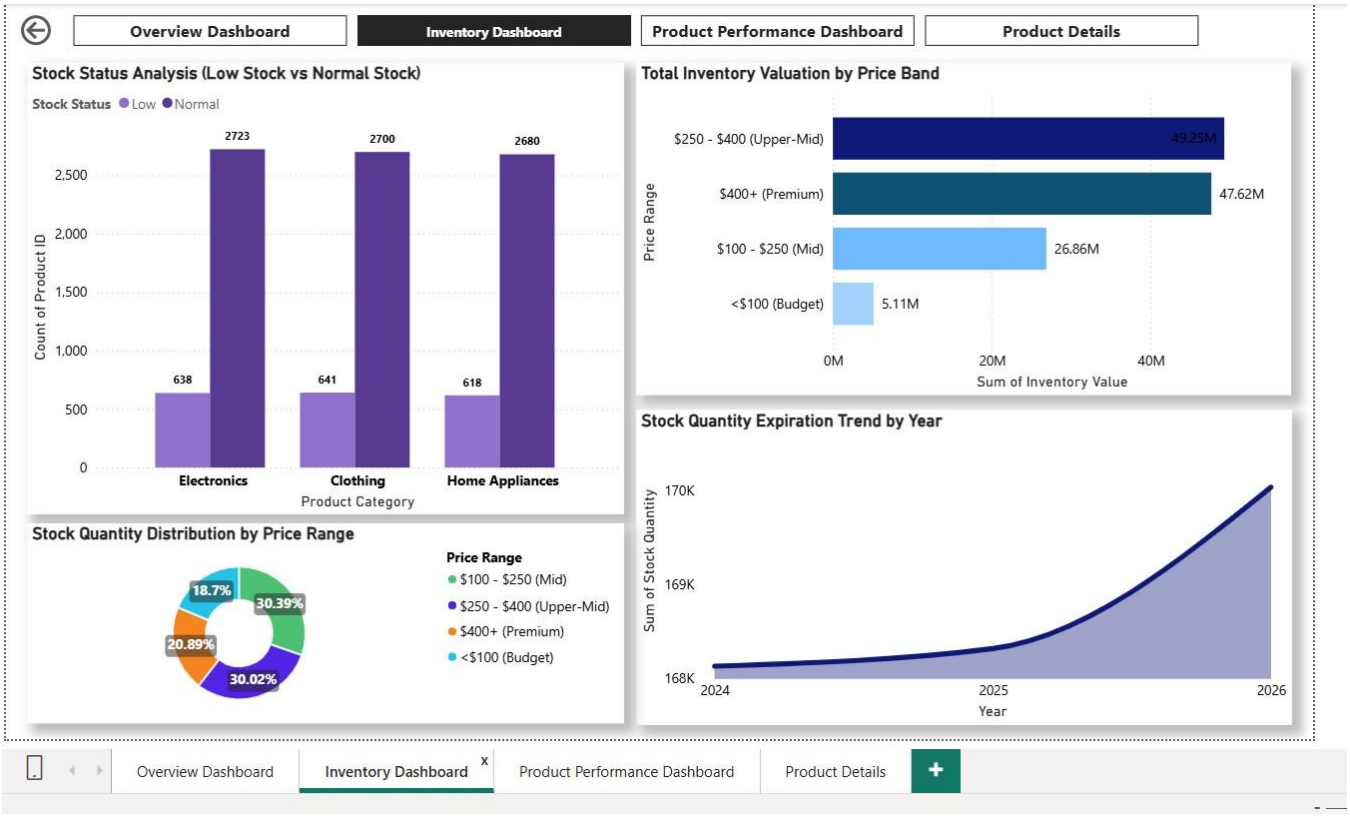


Figure 6.1 – Inventory Dashboard

The inventory dashboard focuses on stock condition and the value of products held by the business. The low-stock comparison is fairly even across the three categories: Electronics has 638 low-stock products, Clothing has 641 and Home Appliances has 618. This suggests that replenishment should not be treated as a problem belonging to only one category.

6.1 Inventory Value by Price Range

The $250–$400 price range has the highest inventory value at $49.25M. The $400+ range follows with $47.62M. The $100–$250 range contributes $26.86M, while products below $100 account for $5.11M. The first two ranges together contain $96.87M of inventory value. This is more than three quarters of the reported $128.83M total. Therefore, high-priced stock should receive close attention because slow movement in these products can affect a large amount of working capital.

6.2 Expiry Trend

The expiry trend increases from around 168K units in 2024 to about 170K units in 2026. The increase is not extremely large, but it is important because it represents stock that may become more difficult to sell as it approaches expiry.

6.3 Stock Quantity Distribution by Price Range

The donut chart on the Inventory Dashboard breaks down stock quantity, rather than stock value, by price band: the $100-$250 (Mid) band holds 30.39% of stock quantity, the $250-$400 (Upper-Mid) band holds 30.02%, the $400+ (Premium) band holds 20.89%, and the under-$100 (Budget) band holds 18.7%. Comparing this with the value-based split in section 6.1 shows an important pattern: the Premium and Upper-Mid bands hold a much larger share of inventory value ($96.87M, or over 75% of total value) than of stock quantity (around 51% of units combined). This means a relatively small proportion of physical units accounts for most of the money tied up in inventory, which reinforces the recommendation to prioritise the higher-priced bands when planning stock reviews.

7. Product Performance Analysis

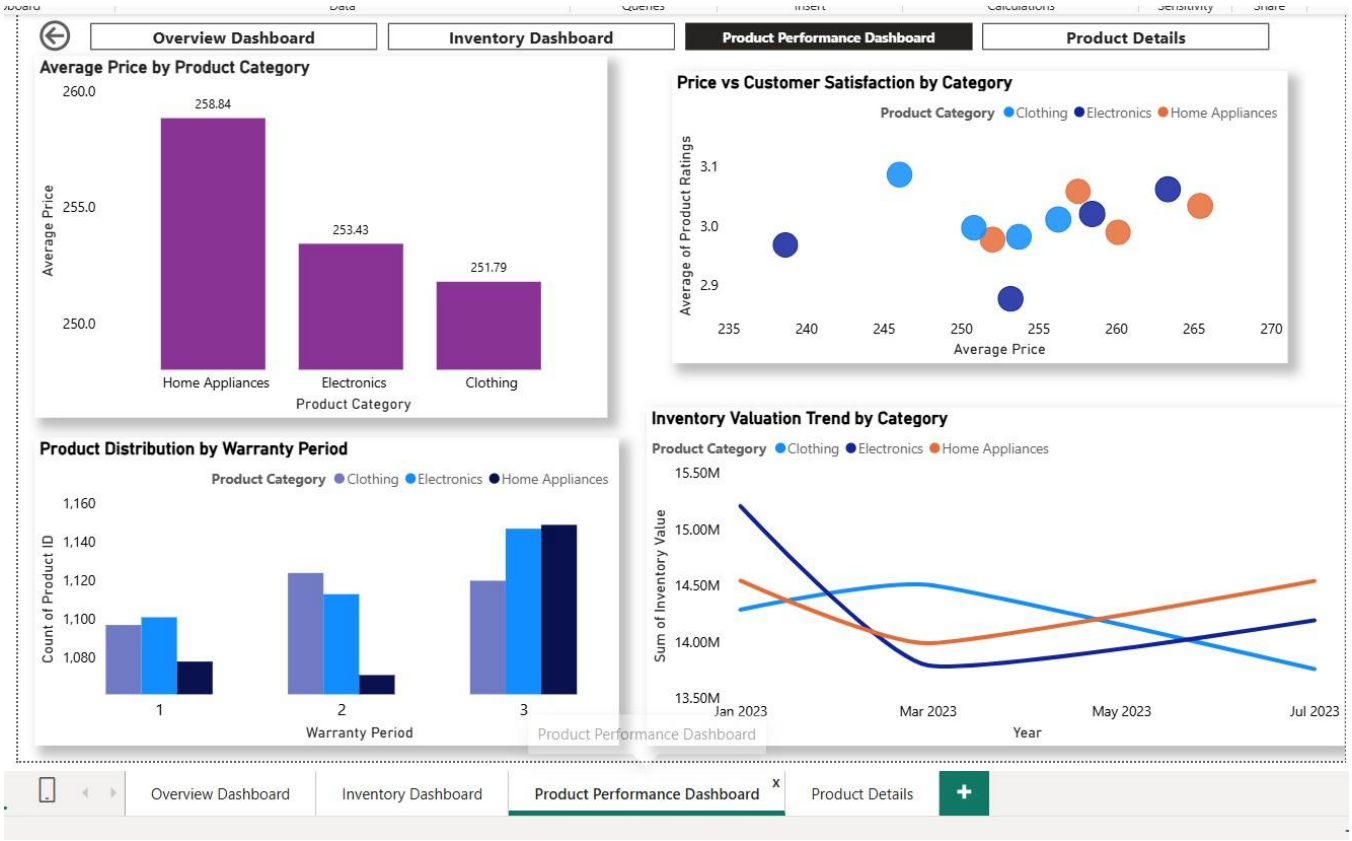


Figure 7.1 – Product Performance Dashboard

The product-performance page compares price, category, warranty and customer ratings. Home Appliances has the highest average price at $258.84. Electronics follows at $253.43, while Clothing has an average price of $251.79. The difference between categories is relatively small, so average price alone does not clearly separate the categories.

7.1 Price and Rating

The scatter plot gives an important result. Product ratings stay within a narrow range of roughly 2.88 to 3.09 even when the average prices change. A Clothing point combines a relatively low price with a high rating, while an Electronics point has the lowest rating.

Based on this dashboard, there is no strong indication that higher prices automatically lead to higher customer ratings. For the business, this means that improving product quality and customer service may be more useful than depending on price increases to improve satisfaction.

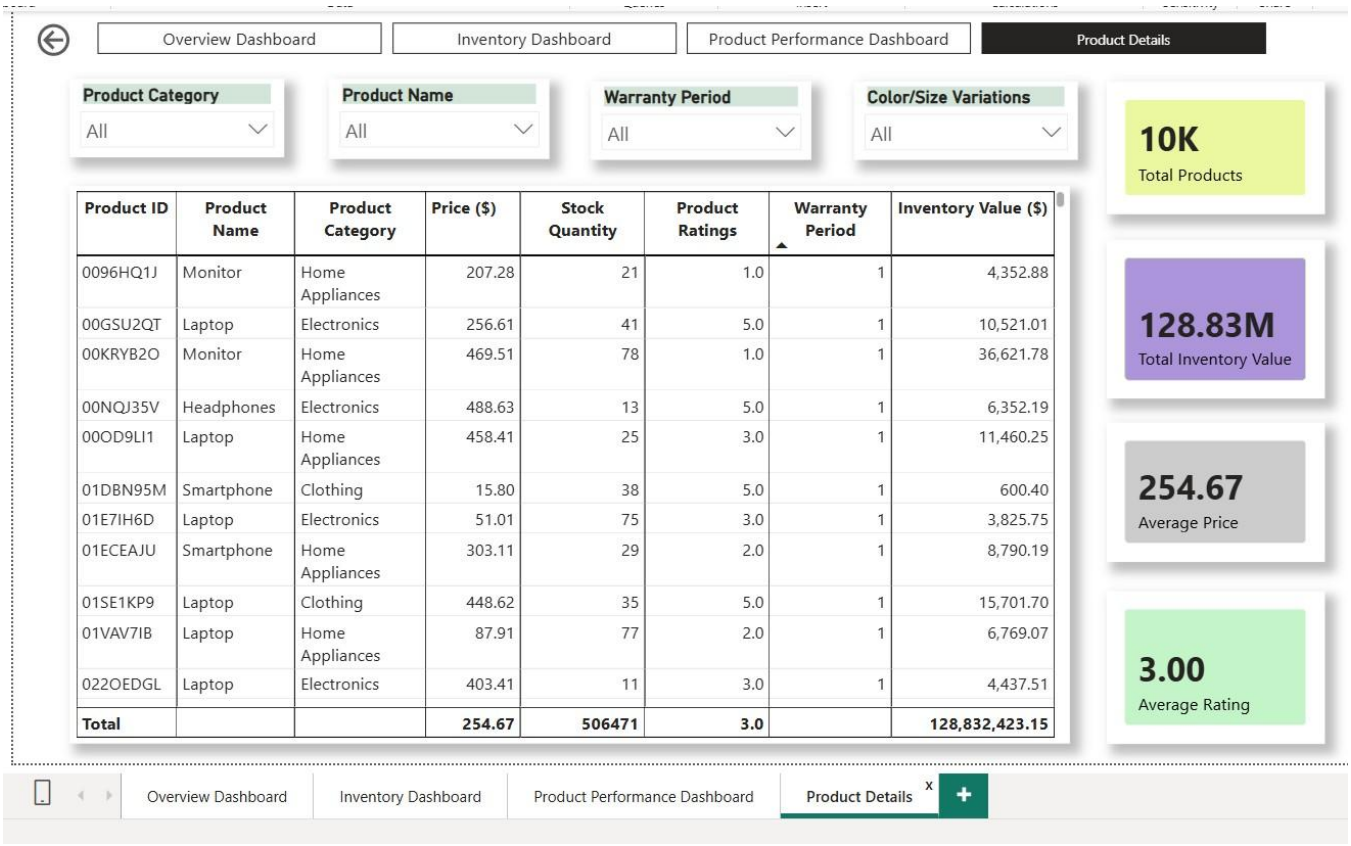
7.2 Warranty Information

The dashboard also includes the distribution of products by warranty period. This gives the business another way to compare product groups and can be useful when reviewing customer coverage and product value. The warranty slicer allows the user to focus on a particular warranty period when required. Warranty periods 1, 2 and 3 each hold a broadly similar product count within every category (roughly 1,080-1,160 products per category per warranty period), so warranty coverage does not appear to be concentrated in a single category. Read together with the price comparison earlier in this section, Home Appliances combines the highest average price ($258.84) with a warranty distribution similar to the other two categories, so customers paying the most are not being offered noticeably longer warranty coverage in return - this is a point the business may want to review.

7.3 Inventory Value Trend by Category

The line chart on the Product Performance Dashboard tracks inventory value by category from January to July. Electronics starts as the highest-value category but declines steadily over the period. Home Appliances dips slightly in the early months before recovering and ending as the strongest of the three by July, while Clothing rises through the middle of the period before falling back towards its starting level. Home Appliances is therefore the best-performing category overall in this report: it combines the highest average price, a recovering/upward inventory-value trend, and a low-stock share in line with the other categories.

8. Drill-through Analysis



Product ID	Product Name	Product Category	Price ($)	Stock Quantity	Product Ratings	Warranty Period	Inventory Value ($)
0096HQ1J	Monitor	Home Appliances	207.28	21	1.0	1	4,352.88
00GSU2QT	Laptop	Electronics	256.61	41	5.0	1	10,521.01
00KRYB2O	Monitor	Home Appliances	469.51	78	1.0	1	36,621.78
00NQJ35V	Headphones	Electronics	488.63	13	5.0	1	6,352.19
00OD9LI1	Laptop	Home Appliances	458.41	25	3.0	1	11,460.25
01DBN95M	Smartphone	Clothing	15.80	38	5.0	1	600.40
01E7IH6D	Laptop	Electronics	51.01	75	3.0	1	3,825.75
01ECEAJU	Smartphone	Home Appliances	303.11	29	2.0	1	8,790.19
01SE1KP9	Laptop	Clothing	448.62	35	5.0	1	15,701.70
01VAV7IB	Laptop	Home Appliances	87.91	77	2.0	1	6,769.07
022OEDGL	Laptop	Electronics	403.41	11	3.0	1	4,437.51
Total			**254.67**	**506471**	**3.0**		**128,832,423.15**

Figure 8.1 – Product Details Drill-through

The drill-through page is useful after an issue has been identified in the summary dashboard. It lists product-level information including product ID, product name, category, price, stock quantity, rating, warranty period and inventory value.

For example, if a category shows a high number of low-stock products, the user does not need to return to the raw CSV file. The drill-through function can be used to inspect the individual products in that selection. This makes the dashboard more useful for actual decision-making.

8.1 Interactivity in the Report

- Category slicer – focuses the analysis on a selected category.
- Product slicer – allows an individual product to be selected.
- Warranty slicer – compares products according to warranty period.
- Colour/Size slicer – narrows the analysis using available variations.
- Drill-through – moves from summary analysis to product-level information.

9. Business Insights and Recommendations

9.1 Main Findings

1. **Inventory value is concentrated in expensive stock.** The $250–$400 and $400+ ranges together account for $96.87M, so these products should be monitored carefully.

2. **Low-stock pressure is spread across categories.** The low-stock counts are 638, 641 and 618 for Electronics, Clothing and Home Appliances respectively. The issue is therefore not limited to one category.

3. **Home Appliances has the highest average price.** Its average price is $258.84, compared with $253.43 for Electronics and $251.79 for Clothing.

4. **Price does not clearly explain customer rating.** Ratings remain around 2.88–3.09 across the displayed categories and prices.

5. **Expiry-related stock is increasing.** The trend moves from about 168K units in 2024 towards 170K units in 2026, so ageing stock should be monitored.

6. **Home Appliances is the best-performing category.** It records the highest average price ($258.84), and the Inventory Value Trend chart (section 7.3) shows it recovering to become the strongest category by value by the end of the period, unlike Electronics which trends downward. Its low-stock count (618) is also the lowest of the three categories.

7. **Individual high-value products can be pinpointed through the drill-through page.** For example, product 00KRYB2O (a Monitor in Home Appliances) has an inventory value of $36,621.78 from just 78 units in stock, far above most other rows in the same table. Reviewing the product-level table regularly, rather than only the category-level totals, allows the business to catch concentration risk in specific highvalue products before it becomes a large write-off.

9.2 Recommendations

- Set a regular replenishment review because low-stock products appear in all three categories.
- Give additional attention to the $250–$400 and $400+ groups because they contain most of the inventory value.
- Identify ageing products early and consider stock rotation, promotions or clearance before expiry becomes a major problem.

- Do not use price as the only method for improving customer satisfaction; review product quality and service-related factors as well.
- Use the drill-through page to identify individual products behind unusual KPI or chart results.
- Continue using the dashboard filters during periodic inventory reviews so that decisions can be made for specific categories, products or warranty groups.
- Use Home Appliances' recovering inventory-value trend and low stock-shortage share as a model, and review what is driving Electronics' declining trend before it affects revenue.
- Review whether warranty length should scale with price, since the highest-priced category (Home Appliances) does not currently receive noticeably longer warranty coverage than the other categories.

These recommendations are based on the dashboard results and the business-insight areas specified in the assessment guideline.

10. Conclusion

This project shows how Power BI can be used to turn a product dataset into a more useful business report. The dashboard brings together inventory, price, rating, warranty and expiry-related information and presents it through KPIs, charts, slicers and a drill-through page.

The analysis found that most inventory value is held in the higher price ranges, while low-stock products are spread across the three categories. Customer ratings are fairly close across the price levels, so higher prices do not appear to guarantee better satisfaction. The expiry trend also shows that ageing inventory needs attention.

Overall, the dashboard can support three main decisions: when stock needs replenishment, where the largest amount of inventory money is concentrated, and which ageing products may require action. This makes the report useful not only as a visualisation exercise but also as a basic business-analysis tool.

11. References

[2] Microsoft Power BI Documentation
https://drive.google.com/file/d/1u6isIa83dCgtbwikFaY9_rpT2eTA2UrL/view?usp=sharing